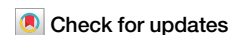




Restoring Heisenberg scaling in time via autonomous quantum error correction



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We establish a sufficient condition under which autonomous quantum error correction (AutoQEC) can effectively restore Heisenberg scaling (HS) in quantum metrology. Specifically, we show that if all Lindblad operators associated with the noise commute with the signal Hamiltonian and a particular constrained linear equation admits a solution, then an ancilla-free AutoQEC scheme with finite R (where R represents the ratio between the engineered dissipation rate for AutoQEC and the noise rate) can approximately preserve HS with a desired small additive error $\epsilon > 0$ over any time interval $0 \leq t \leq T$. We emphasize that the error scales as $\epsilon = O(\kappa T/R^c)$ where c is a positive integer and κ is the noise rate, indicating that the required R decreases significantly with increasing c to achieve a desired error. Furthermore, we discuss that if the sufficient condition is not satisfied, logical errors may be induced that cannot be efficiently corrected by the canonical AutoQEC framework. Finally, we numerically verify our analytical results by employing concrete examples of phase estimation under dephasing noise.

Quantum metrology investigates how quantum resources can enhance the precision of signal estimation. Over the past years, numerous studies have shown that quantum coherence and entanglement can potentially achieve higher estimation precision compared to classical methods^{1–5}. These advantages have been explored in a wide range of applications, including gravitational wave detection, interferometry, magnetometry, and atomic clocks^{6–14}. A central goal in quantum metrology is to attain the *Heisenberg scaling* (HS) wherein the variance of the estimator scales as $1/T^2$ with total sensing time T , representing an ultimate quantum-enhanced precision limit². However, in practice, achieving HS is often hindered by decoherence induced by interactions with the environment, limiting the attainable precision below HS^{15–18}.

To tackle these challenges, quantum error correction (QEC)—primarily developed within the context of quantum computation—has been employed to restore HS^{19–26}. It has been shown that as long as the noise remains distinguishable from the signal—formally characterized by the *Hamiltonian-not-in-Lindblad-span* (HNLS) condition—QEC protocols capable of preserving HS can be constructed^{22–24}. The necessity of HNLS condition—which does not arise in the context of quantum computation—stems from the fundamental constraint that the signal Hamiltonian is fixed and cannot be arbitrarily manipulated in response to the structure of the noise. However, even when the HNLS condition is satisfied, the QEC protocols capable of restoring HS typically require continuous monitoring and active feed-forward processes

(referred to as *ideal QEC* below), which impose substantial resource demands and present significant experimental challenges.

Autonomous quantum error correction (AutoQEC) can be a promising candidate to address these challenges. AutoQEC leverages engineered dissipation to protect quantum states from noise without continuous monitoring or active feed-forwards operations, thereby potentially avoiding the substantial resource requirements of ideal QEC^{27–30}. By exploiting engineered dissipation, AutoQEC suppresses the decoherence rate κ to $O(\kappa/R^c)$ at the logical level (see Fig. 1a), where $R \gg 1$ represents the ratio between the engineered dissipation rate and the decoherence rate, and c is a positive integer referred to as the *order of AutoQEC* that is determined by an AutoQEC code³¹. In quantum computation, AutoQEC has demonstrated its capability to efficiently suppress errors, both theoretically^{32–37} and in experimental implementations^{38–41}. However, its capability to restore HS remains unclear, as the application of AutoQEC to metrology faces two fundamental challenges: the signal Hamiltonian is fixed, and R is finite in its practical implementation. For instance, ref. 42 explored the application of AutoQEC to quantum metrology; however, both the signal Hamiltonian and the environmental noise introduced logical errors, ultimately preventing the restoration of HS. Furthermore, it is worth emphasizing that, while AutoQEC codes for metrology have been proposed in specific settings^{42,43}, a general and systematic construction of an AutoQEC code tailored for metrology has remained unexplored.

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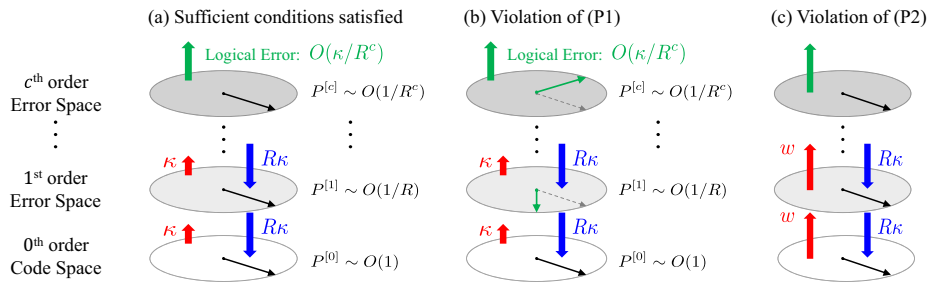


Fig. 1 | Schematic illustration of AutoQEC dynamics under different conditions.

Each panel depicts the evolution of quantum states across the code space and higher-order error spaces. **a** The engineered dissipation (blue thick arrows) effectively counteracts natural dissipation (red thick arrows), leading to a suppression of the probability of occupying the $0 \leq n \leq c$ -th order error space, scaling as $P^{[n]} \sim O(1/R^n)$. Consequently, the decoherence (induced by the natural dissipation) rate is suppressed to $O(\kappa/R^c)$ at the logical level, also relevant to AutoQEC dynamics in

quantum computation. **b** Phase differences between subspaces introduce logical errors, represented by the green thin arrows in the error spaces. As a result, even though the decoherence rate is suppressed to $O(\kappa/R^c)$, one may not achieve AutoQEC up to order c . **c** In addition to natural dissipation, $\hat{U}(wt) := e^{-i\hat{H}wt}$ induces additional errors that transfer states into higher-order error spaces, indicated by w -induced red thick arrows. These errors accumulate more rapidly than those caused by natural dissipation, in particular when $w \gg \kappa$.

In this work, we establish a sufficient condition under which AutoQEC can effectively restore HS in quantum metrology. Specifically, we show that if all Lindblad operators of the noise commute with the signal and a constrained linear equation (introduced in Theorem 1) has a solution, then it is possible to construct an ancilla-free AutoQEC code with finite R , tailored for quantum metrology that efficiently preserves HS within small additive error $\epsilon > 0$ over any given time interval $0 \leq t \leq T$. (In the remainder of the paper, we refer to this approximate preservation of HS simply as HS.) To the best of our knowledge, this constitutes the first general AutoQEC code explicitly tailored for quantum metrology, and importantly, the code does not require a noiseless ancilla. In addition, the error scales as $\epsilon = O(\kappa T/R^c)$, indicating that, to achieve the targeted error ϵ , the required R decreases substantially with the AutoQEC order c . These features can be advantageous for practical implementations, as they both eliminate the need for noiseless ancillas and alleviate the experimental challenge of realizing very large R ³¹.

We also discuss potential challenges arising when the sufficient condition is not satisfied. In such cases, the signal Hamiltonian may induce logical errors that cannot be efficiently corrected by the canonical AutoQEC. To validate our results, we apply AutoQEC to two phase-estimation scenarios: (i) a 3-qubit system governed by the signal Hamiltonian $\hat{H} = \sum_{i=1}^3 \hat{Z}_i$ under correlated dephasing noise, and (ii) a 5-qubit system with $\hat{H} = \prod_{i=1}^5 \hat{Z}_i$ under local dephasing noise. Our numerical results show that when our sufficient condition is satisfied, AutoQEC successfully restores HS.

Results

Autonomous quantum error correction in computation

Before introducing our main results, we briefly review AutoQEC in the context of quantum computation³¹. In the presence of time-independent Markovian noise, the dynamics of the system are governed by the Lindblad master equation

$$\frac{d\hat{\rho}(t)}{dt} = -i[w\hat{H}, \hat{\rho}(t)] + \kappa\tilde{\mathcal{L}}_n[\hat{\rho}(t)] + R\kappa\tilde{\mathcal{L}}_E[\hat{\rho}(t)]. \quad (1)$$

Here, we have that (1) $\hat{H}$ is the control Hamiltonian for desired logical operations, and w is a parameter that governs the corresponding unitary evolution. (2) $\kappa\tilde{\mathcal{L}}_n[\hat{\rho}(t)] := \sum_{a>0} D[\hat{L}_{n,a}][\hat{\rho}(t)]$ is the natural dissipation which corresponds to noise, with dissipation rate κ , $\hat{L}_{n,a}$ are the Lindblad operators, and $D[\hat{A}][\hat{\rho}(t)] := \hat{A}\hat{\rho}(t)\hat{A}^\dagger - \frac{1}{2}\{\hat{A}^\dagger\hat{A}, \hat{\rho}(t)\}$. (3) $R\kappa\tilde{\mathcal{L}}_E[\hat{\rho}(t)] := \sum_{b>0} D[\hat{L}_{E,b}][\hat{\rho}(t)]$ is the engineered dissipation which performs AutoQEC, where $R\kappa$ is the engineered dissipation rate, and $\hat{L}_{E,b}$ are the corresponding Lindblad operators.

The purpose of AutoQEC is to restore the desired target state $\hat{U}_0(wt)\hat{\rho}_0\hat{U}_0^\dagger(wt)$ where $\hat{U}_0(wt) := e^{-i\hat{H}_0wt}$, by applying engineered dissipation $\tilde{\mathcal{L}}_E$ and control Hamiltonian $\hat{H}$. Here, $\hat{\rho}_0$ is the initial state that lies within the code space $\mathcal{C}$, and $\hat{H}_0$ is the logical Hamiltonian satisfying $\hat{H}_0 = \hat{\Pi}_C\hat{H}_0\hat{\Pi}_C$, where $\hat{\Pi}_C$ is the projection operator onto the code $\mathcal{C}$.

Next, we introduce the definition of AutoQEC up to order c ³¹:

Definition 1. $\tilde{\mathcal{L}}_E$ and $\hat{H}$ are said to perform AutoQEC up to order c with respect to the code space $\mathcal{C}$, the target logical Hamiltonian $\hat{H}_0$ and the natural dissipation $\tilde{\mathcal{L}}_n$, iff there exist two positive constants $M, w_0 > 0$ such that for any initial condition $\hat{\rho}_0 \in \mathcal{C}$ one has

$$\|\tilde{\mathcal{P}}_E[\hat{\rho}(t)] - \hat{U}_0(wt)\hat{\rho}_0\hat{U}_0^\dagger(wt)\| \leq M\|\hat{\rho}(0)\| \frac{\kappa t}{R^c} \quad (2)$$

$\forall t, R \geq 0$ and with w satisfying $|w| \leq w_0 R$.

Here, $\tilde{\mathcal{P}}_E := \lim_{u \rightarrow \infty} e^{\tilde{\mathcal{L}}_E u}$ is a CPTP projector stabilizing the code space^{31,44}. Based on Definition 1, we now state the main result of ref. 31 (detailed definitions of the relevant terms in the below lemma are provided further below):

Lemma 1. If the Knill-Laflamme condition is satisfied for the error set $\mathcal{E}^{[c]}$, then AutoQEC up to order c can be performed by applying the following $\tilde{\mathcal{L}}_E$ and $\hat{H}$:

$$\tilde{\mathcal{L}}_E = \sum_{n=1}^c \sum_{i_n=1}^{p_n} D[\hat{L}_{E,i_n}^{[n]}] + \sum_{q=1}^{q_{\max}} D[\hat{L}_{E,q}^{[\text{res}]}], \quad (3)$$

$$\hat{H} = \sum_{j,k=0}^{d_c-1} \sum_{n=0}^c \sum_{i_n=1}^{p_n} \langle \mu_j | \hat{H}_0 | \mu_k \rangle \langle \mu_{j,i_n}^{[n]} | \mu_{k,i_n}^{[n]} \rangle, \quad (4)$$

where $\hat{L}_{E,i_n}^{[n]} = \sum_{j=0}^{d_c-1} |\mu_j\rangle \langle \mu_{j,i_n}^{[n]}|$ for $1 \leq n \leq c$ and $1 \leq i_n \leq p_n$, and $\hat{L}_{E,q}^{[\text{res}]} = |\Phi_q\rangle \langle \Phi_q|$ for $1 \leq q \leq q_{\max}$. Here, $|\Phi_q\rangle$ can be chosen among any normalized states in $\mathcal{C}$.

Lemma 1 contains the followings:

(1) Error set: Define the zeroth-order error set $\mathcal{E}^{[0]} := \{\hat{I}\}$, the first-order error set $\mathcal{E}^{[1]} := \{\hat{L}_{n,1}, \hat{L}_{n,2}, \dots, \hat{L}_{n,p_n}\}$, and the error that corresponds to the no-jump evolution $\hat{B} := \sum_{a=1}^{N_n} \hat{L}_{n,a}^\dagger \hat{L}_{n,a}$. We can recursively define the n -th order error set as

$$\begin{aligned} \mathcal{E}^{[n]} &:= \{\hat{L}_{n,a} \hat{E}_l^{[n-1]} | \hat{E}_l^{[n-1]} \in \mathcal{E}^{[n-1]} \forall a\} \\ &\cup \{\hat{B} \hat{E}_l^{[n-2]} | \hat{E}_l^{[n-2]} \in \mathcal{E}^{[n-2]}\}. \end{aligned} \quad (5)$$

Finally, we define the error set up to order n as

$$\mathcal{E}^{[\sim n]} := n \cup \mathcal{E}^{[k]}. \quad (6)$$

(2) Basis: Let us denote the basis of the code space $\mathcal{C}$ as $\{|\mu_0\rangle, |\mu_1\rangle, \dots, |\mu_{d_c-1}\rangle\}$. By using the Gram-Schmidt algorithm, the following orthonormal basis for the correctable error spaces can be defined as

$$\begin{aligned} & \mu_i^{[\sim c]} \\ & := \left\{ |\mu_i\rangle = |\mu_{i,1}^{[0]}\rangle, |\mu_{i,1}^{[1]}\rangle, \dots, |\mu_{i,p_1}^{[1]}\rangle, \dots, |\mu_{i,1}^{[c]}\rangle, \dots, |\mu_{i,p_c}^{[c]}\rangle \right\} \\ & = \text{G.S.} \left(\left\{ |\mu_i\rangle, \hat{E}_1^{[1]}|\mu_i\rangle, \dots, \hat{E}_{[\sim c]}^{[c]}|\mu_i\rangle \right\} \right). \end{aligned} \quad (7)$$

Here, the vectors $\{|\mu_{i,n}^{[n]}\rangle\}_{i,n=1}^{p_n}$ represent orthonormal vectors in n -th order correctable error space generated from the code word $|\mu_i\rangle \in \mathcal{C}$. Furthermore, the residual space is denoted as $\mathcal{R} := \left(\text{span} \left\{ \bigcup_{i=0}^{d_c-1} \mu_i^{[\sim c]} \right\} \right)^\perp = \text{span} \{|\phi_1\rangle, |\phi_2\rangle, \dots, |\phi_{q_{\max}}\rangle\}$ where $q_{\max} = d_{\mathcal{H}} - d_{\mathcal{C}} \sum_{n=0}^c p_n$, where $d_{\mathcal{H}}$ and $d_{\mathcal{C}}$ is dimension of total Hilbert and code space, respectively.

Application of AutoQEC to metrology. We first clarify the definition of HS. Consider a scenario where a parameter w to be estimated is imprinted on a probe state $\hat{\rho}_0$ via unitary $\hat{U}(wt) = e^{-i\hat{H}wt}$, resulting in the ideal signal state $\hat{\rho}_{\text{id}}(t) := \hat{U}(wt)\hat{\rho}_0\hat{U}^\dagger(wt)$ where t is sensing time. The estimation error of w is fundamentally lower-bounded by the inverse of the quantum Fisher information (QFI) $F[\hat{\rho}_{\text{id}}(t)] = 4t^2 \left(\text{Tr}[\hat{H}^2\hat{\rho}_0] - \text{Tr}[\hat{H}\hat{\rho}_0]^2 \right)^{45-47}$. Unless the initial probe $\hat{\rho}_0$ is an eigenstate of $\hat{H}$, the corresponding QFI scales as $F[\hat{\rho}_{\text{id}}(t)] = \Theta(t^2)$, which is referred to as HS.

We emphasize that there is a fundamental distinction between the application of AutoQEC in the computation and metrology: In metrology, the signal Hamiltonian $\hat{H}$ is predetermined, which is thus beyond our control. Consequently, satisfying the Knill-Laflamme condition does not necessarily guarantee AutoQEC up to order c in metrology, as the validity of Lemma 1 depends on the specific design of the Hamiltonian. This distinction underscores the unique challenges of applying AutoQEC in metrological scenarios and suggests that a stricter condition than the Knill-Laflamme criterion may be required. Motivated by this, we investigate conditions under which the application of AutoQEC restores HS.

Before presenting our sufficient condition, we introduce matrices that simplify the description of our results. Let us express the Hamiltonian $\hat{H}$ in diagonal form as

$$\hat{H} = \sum_{i=0}^{d-1} h_i \left(\sum_{l=1}^{N_i} |h_i^{(l)}\rangle \langle h_i^{(l)}| \right), \quad (8)$$

where $\{h_i\}_{i=0}^{d-1}$ are distinct eigenvalues, and $\{|h_i^{(k)}\rangle\}_{k=1}^{N_i}$ the corresponding eigenvectors. Next, we define the following operator set:

$$\mathcal{K}^{[\sim c]} := \{\hat{E}_a^\dagger \hat{E}_b | \hat{E}_a, \hat{E}_b \in \mathcal{E}^{[\sim c]} \forall a, b\}. \quad (9)$$

Notably, the Knill-Laflamme condition can be expressed as

$$\langle \mu_i | \hat{K}_k | \mu_j \rangle = \sigma_k \delta_{ij} \forall 1 \leq k \leq |\mathcal{K}^{[\sim c]}|, \quad (10)$$

where $\hat{K}_k \in \mathcal{K}^{[\sim c]}$ for all k , and σ_k are constants associated with $\hat{K}_k$. Next, using $\mathcal{K}^{[\sim c]}$ and the eigenvectors of $\hat{H}$, we define $|\mathcal{K}^{[\sim c]}| \times N_i$ matrix $\mathbf{A}_i^{[\sim c]}$, with elements

$$[\mathbf{A}_i^{[\sim c]}]_{kl} := \langle h_i^{(l)} | \hat{K}_k | h_i^{(l)} \rangle \forall k, l. \quad (11)$$

Based on $\mathbf{A}_i^{[\sim c]}$, we introduce the following Theorem:

Theorem 1. Suppose the following two conditions hold: (T1) $[\hat{H}, \hat{L}_{n,a}] = 0 \forall a$, and (T2) $\exists i, j \neq i$, along with two probability vectors $\mathbf{p}_i \in \mathbb{R}_{\geq 0}^{N_i}, \mathbf{p}_j \in \mathbb{R}_{\geq 0}^{N_j}$ such that $\mathbf{A}_i^{[\sim c]} \cdot \mathbf{p}_i = \mathbf{A}_j^{[\sim c]} \cdot \mathbf{p}_j$.

Then, for any given time $T \geq 0$, and an arbitrarily small error $\epsilon > 0$, there exists an ancilla-free AutoQEC scheme with a finite R such that

$$\forall 0 \leq t \leq T : F[\hat{\rho}(t)] \geq (h_i - h_j)^2 t^2 - \epsilon, \quad (12)$$

where $\hat{\rho}(t)$ is the AutoQEC state, and $F[\hat{\rho}]$ is the QFI of $\hat{\rho}$ with respect to w . The relationship between R, κ, T , and ϵ here is given by $\epsilon = O(\kappa T/R^c)$.

Proof sketch [see Methods or Supplementary Material (SM) Section S1 for details]. We consider the following two-dimensional code $\mathcal{C}$ where each codeword is composed of a linear combination of eigenvectors of $\hat{H}$ corresponding to the same eigenvalue:

$$|\mu_0\rangle := \sum_{k=1}^{N_i} \sqrt{[\mathbf{p}_i]_k} |h_i^{(k)}\rangle, \quad |\mu_1\rangle := \sum_{k=1}^{N_j} \sqrt{[\mathbf{p}_j]_k} |h_j^{(k)}\rangle. \quad (13)$$

First, we demonstrate that for this code, (T1) guarantees $\langle \mu_0 | \hat{K}_k | \mu_1 \rangle = 0$ without the need for any ancilla, thereby aligning with previous ancilla-free ideal QEC results²² and (T2) ensures that there exist $\mathbf{p}_i$ and $\mathbf{p}_{j \neq i}$ such that $\langle \mu_0 | \hat{K}_k | \mu_0 \rangle = \langle \mu_1 | \hat{K}_k | \mu_1 \rangle \forall \hat{K}_k \in \mathcal{K}^{[\sim c]}$, i.e., the Knill-Laflamme condition holds for the error set $\mathcal{E}^{[\sim c]}$.

Next, we consider the initial probe state

$$\hat{\rho}_0 := |\psi_0\rangle \langle \psi_0|, \quad |\psi_0\rangle := \frac{|\mu_0\rangle + |\mu_1\rangle}{\sqrt{2}}. \quad (14)$$

For the noiseless scenario, $\hat{\rho}_0$ achieves HS, yielding the corresponding QFI $F[\hat{\rho}_{\text{id}}(t)] = (h_i - h_j)^2 t^2$ where the imprinted state is given by

$$\hat{\rho}_{\text{id}}(t) := e^{-i\hat{H}wt} \hat{\rho}_0 e^{i\hat{H}wt}. \quad (15)$$

Since $|\mu_0\rangle$ and $|\mu_1\rangle$ are eigenstates of $\hat{H}$, the evolution can be equivalently expressed as

$$\hat{\rho}_{\text{id}}(t) = e^{-i\hat{H}_0 wt} \hat{\rho}_0 e^{i\hat{H}_0 wt}, \quad (16)$$

with the effective (logical) Hamiltonian

$$\hat{H}_0 := h_i |\mu_0\rangle \langle \mu_0| + h_j |\mu_1\rangle \langle \mu_1|. \quad (17)$$

We then establish that AutoQEC up to order c , as defined in Eq. (2), is achievable. Moreover, Eq. (2) implies

$$|F[\tilde{\mathcal{P}}_E[\hat{\rho}(t)] - F[\hat{\rho}_{\text{id}}(t)]| \leq \epsilon. \quad (18)$$

Here, since $\tilde{\mathcal{P}}_E$ is a CPTP map, the data processing inequality of the QFI ensures⁴⁸

$$F[\tilde{\mathcal{P}}_E[\hat{\rho}(t)]] \leq F[\hat{\rho}(t)] \leq F[\hat{\rho}_{\text{id}}(t)]. \quad (19)$$

Combining Eqs. (18) and (19) then yields Eq. (12). Finally, we emphasize that the final CPTP map $\tilde{\mathcal{P}}_E$ is not required to restore HS. Indeed, by the data-processing inequality in Eq. (19), any such CPTP map does not increase the QFI. Therefore, it is unnecessary to apply $\tilde{\mathcal{P}}_E$ to restore HS by the AutoQEC scheme. $\square$

We highlight several important aspects of Theorem 1. First, our code does not require any noiseless ancilla. Second, the error scales as $\epsilon = O(\kappa T/R^c)$, implying that increasing the AutoQEC order c enables a fixed target

accuracy ϵ to be achieved with a smaller engineered dissipation strength R . This is particularly advantageous in practical regimes where achieving large R is subject to stringent constraints. At the same time, it is important to emphasize that achieving higher-order AutoQEC generally requires a more intricate recovery structure; in the canonical construction, this manifests as an increased number of recovery Lindblad operators $\hat{L}_{E,i,n}^{[n]}$, as indicated by Eq. (3) [see SM Section S5 for details]. However, rather than constituting a limitation, this reflects a favorable and systematic trade-off: higher-order AutoQEC reduces the required dissipation strength while affording additional flexibility in the recovery design. This interplay between dissipation resources and recovery complexity provides a practical pathway toward efficient implementations.

Third, identifying $(\mathbf{p}_i, \mathbf{p}_j)$, which corresponds to constructing an AutoQEC code tailored for metrology, reduces to solving a constrained linear feasibility problem. This can be addressed in a systematic manner using *linear programming* methods, and can be efficiently performed, particularly for modest system sizes [see SM Sec. S2]. Third, if our sufficient condition is satisfied, the HNLS condition, which is the necessary and sufficient condition for ideal QEC to fully restore HS²⁴, is automatically satisfied [see SM Section S1]. This follows from the fact that the code defined in Eq. (13) satisfies the Knill-Laflamme condition and ensures $\hat{\Pi}_C \hat{H} \hat{\Pi}_C \neq h \hat{\Pi}_C$ where h is a constant, which implies that HNLS is satisfied^{24,49}.

Finally, we discuss the scenarios where the sufficient condition in Theorem 1 is not satisfied. In Theorem 2 of SM Section S3. A, we show that even if the sufficient condition does not hold, AutoQEC can restore HS in the infinite R limit and with the use of a noiseless ancilla, provided that the HNLS condition is satisfied. However, for finite R , it is not guaranteed that an AutoQEC scheme can efficiently restore HS without the satisfaction of the sufficient condition. To gain further insight into this limitation, we now explore cases where the sufficient condition is not satisfied. When the sufficient condition is indeed satisfied and exploiting the code defined in Eq. (13), the signal Hamiltonian $\hat{H}$ exhibits the following two significant properties:

- (P1) $\hat{H}$ commutes with all the errors, i.e., $[\hat{H}, \hat{E}] = 0 \forall \hat{E} \in \mathcal{E}^{[c]}$.
- (P2) $[\hat{\Pi}^{[n]}, \hat{H}] = 0 \forall n$, where $\hat{\Pi}^{[n]} := \sum_{i=0}^1 \sum_{i_n} |\mu_{i,i_n}^{[n]}\rangle \langle \mu_{i,i_n}^{[n]}|$ are projection operators that correspond to an n -th order correctable error space with $0 \leq n \leq c$.

Here, (P1) follows directly from (T1). To show that (P2) is guaranteed under conditions (T1) and (T2), we assume that both conditions are satisfied. We first observe from Eq. (13) that $\hat{\Pi}^{[0]}$ commutes with $\hat{H}$, since each codeword $|\mu_0\rangle$ and $|\mu_1\rangle$ is composed of the eigenvectors of $\hat{H}$ with the same eigenvalue. Furthermore, by construction in Eq. (7), the orthonormal basis for the n -th order correctable error spaces $\{|\mu_{i,i_n}^{[n]}\rangle\}_{i_n=1}^{P_n}$ consists of the linear combinations of

$$\left\{ |\mu_i\rangle, \hat{E}_1^{[1]} |\mu_i\rangle, \dots, \hat{E}_{|\mathcal{E}^{[n]}|}^{[n]} |\mu_i\rangle \right\}. \quad (20)$$

Combined with the fact that $[\hat{H}, \hat{\Pi}^{[0]}] = 0$, this directly guarantees (P2), i.e., $[\hat{H}, \hat{\Pi}^{[n]}] = 0$. As a consequence, violation of the sufficient condition can lead to violation of either (P1) or (P2), and we now address potential challenges. To this end, we consider the engineered dissipation in Eq. (3), a canonical engineered dissipation in AutoQEC. First, consider the case when (P1) is violated; $\hat{U}(wt)$ then possibly acts differently across code and error spaces, resulting in different phase accumulations between these spaces. These phase differences may induce logical errors that cannot be corrected by engineered dissipation [see Fig. 1 (b)]. Second, consider the case when (P2) is violated. In this scenario, $\hat{U}(wt) = e^{-i\hat{H}wt}$ may introduce additional errors that transfer a quantum state to higher-order error spaces. This becomes particularly critical when $w \gg \kappa$ (which is a case of interest in metrology), since then these errors accumulate significantly faster than

those arising from natural dissipation [see Fig. 1 (c)]. Moreover, it is not guaranteed that engineered dissipation can correct such *coherent* errors. Consequently, if our sufficient condition is not satisfied, AutoQEC does not guarantee efficient restoration of HS. More specifically, considering Theorem 2 in SM, if our sufficient condition is violated, we may require substantially larger R to achieve the same error ϵ compared to the case when the sufficient condition is satisfied. In Sec. S3. B of the SM, we numerically implement these scenarios.

Numerical results

For the numerical simulation of our proposed scheme, we consider N -qubit phase estimation in the presence of correlated dephasing noise. The sensing dynamics (in the absence of AutoQEC) is governed by the Lindbladian evolution

$$\frac{d\hat{\rho}}{dt} = -i w [\hat{H}, \hat{\rho}(t)] + \kappa \sum_{i,j=1}^N [\mathbf{C}]_{ij} \left(\hat{Z}_i \hat{\rho} \hat{Z}_j - \frac{1}{2} \{ \hat{Z}_i \hat{Z}_j, \hat{\rho} \} \right), \quad (21)$$

where $\hat{Z}_i$ represents the Pauli-Z operator acting on the i th qubit and $\mathbf{C}$ is the *correlation matrix* that characterizes the spatial structure of the noise. Specifically, the off-diagonal elements $[\mathbf{C}]_{ij}$ quantify the noise correlation between i th qubit and j th qubit.

In the first numerical analysis, we consider the signal Hamiltonian $\hat{H} = \sum_{i=1}^N \hat{Z}_i$. We note that in this setting, the ideal QEC has been shown to restore HS as long as a correlation matrix is not full-rank; equivalently, the HNLS condition is satisfied^{22,24}. However, for AutoQEC, a non-full-rank correlation matrix does not always guarantee fulfilling the conditions of Theorem 1 [see SM Section S1.A for details]. We consider a 3-qubit system with $w = 1$, $\kappa = 0.1$, and

$$\mathbf{C} = \frac{1}{10} \begin{pmatrix} 16 & -4 & -4 \\ -4 & 7 & -5 \\ -4 & -5 & 7 \end{pmatrix}. \quad (22)$$

We note that we carefully chose a correlation matrix that satisfies our sufficient condition. Now the sufficient condition in Theorem 1 is satisfied up to AutoQEC order $c = 1$. Therefore, based on Eq. (13), we consider the following code words:

$$|\mu_0\rangle = (\sqrt{4}|100\rangle + \sqrt{3}|010\rangle + \sqrt{3}|001\rangle)/\sqrt{10}, \quad (23)$$

$$|\mu_1\rangle = (\sqrt{4}|011\rangle + \sqrt{3}|101\rangle + \sqrt{3}|110\rangle)/\sqrt{10}. \quad (24)$$

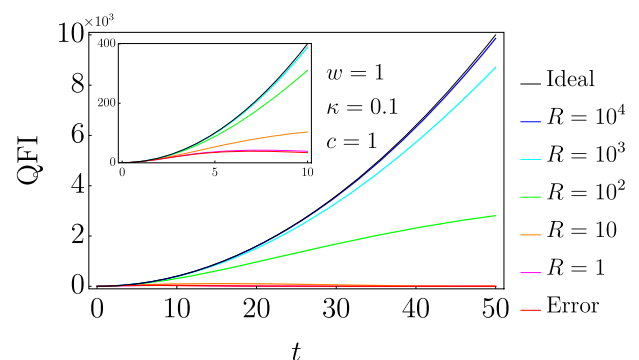


Fig. 2 | QFI under correlated dephasing with AutoQEC for signal Hamiltonian $\hat{H} = \sum_{i=1}^3 \hat{Z}_i$. QFI as a function of sensing time t when the signal Hamiltonian is given by $\hat{H} = \sum_{i=1}^3 \hat{Z}_i$ in the presence of correlated dephasing noise defined in Eq. (22). With the code defined in Eqs. (23) and (24), AutoQEC of order $c = 1$ can be achieved. In the numerics, we consider various values of R , with $w = 1$, $\kappa = 0.1$, and AutoQEC order $c = 1$. The black line (labeled “Ideal”) and the red line (labeled “Error”) represent the QFI of the noiseless case and the case without AutoQEC, respectively. The inset shows a zoomed-in view of the QFI at shorter times t .

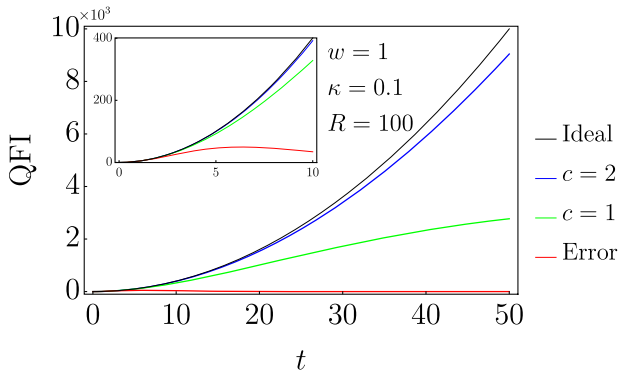


Fig. 3 | QFI under local dephasing with AutoQEC for signal Hamiltonian
 $\hat{H} = \prod_{i=1}^5 \hat{Z}_i$. QFI as a function of time when the signal Hamiltonian is given by $\hat{H} = \prod_{i=1}^5 \hat{Z}_i$, in the presence of local dephasing noise. In this case, by exploiting the repetition code defined in Eqs. (25) and (26), AutoQEC order up to $c = 2$ can be achieved. For the numerics, we consider different AutoQEC orders c , with $R = 100$. For $c = 1$, we consider $|\Phi_q\rangle = (|\mu_0\rangle + |\mu_1\rangle)/\sqrt{2} \forall 1 \leq q \leq q_{\max}$. Other conventions are identical to those of Fig. 2.

In Fig. 2, we show that HS can be efficiently restored by the AutoQEC. Especially, as R increases, HS is more effectively restored.

In the second numerical analysis, we employ the signal $\hat{H} = \prod_{i=1}^N \hat{Z}_i$, in the presence of local dephasing noise where $[C]_{ij} = \delta_{ij}$. This scenario satisfies conditions in Theorem 1 up to AutoQEC order $1 \leq c \leq \lfloor \frac{N-1}{2} \rfloor$ with the code words

$$|\mu_0\rangle = (|+\rangle^{\otimes N} + |-\rangle^{\otimes N})/\sqrt{2}, \quad (25)$$

$$|\mu_1\rangle = (|+\rangle^{\otimes N} - |-\rangle^{\otimes N})/\sqrt{2}. \quad (26)$$

We considered a 5-qubit system with $w = 1$, $\kappa = 0.1$, and $R = 100$. In Fig. 3, we show that HS can be restored by AutoQEC. In particular, a larger c can more efficiently restore HS with the same R . A more detailed analysis of the numerics is found in the SM Section S4.

Discussion

We have established a sufficient condition ensuring that AutoQEC can efficiently restore HS. Notably, our code does not require noiseless ancillas and furthermore enables higher-order AutoQEC, both of which can constitute important practical advantages, especially in settings where noiseless ancillas, large measurement-feedback overhead, or the implementation of sufficiently large recovery rates R are difficult to realize. AutoQEC has already been experimentally demonstrated in quantum computation^{38–41}. In quantum metrology, a theoretical proposal for the implementation of AutoQEC has also been put forward⁴³, and we anticipate future experiments to demonstrate AutoQEC-based quantum estimation, thus overcoming limitations by noise in quantum metrology applications.

We finally emphasize the major differences of our study from ref. 31 which investigates AutoQEC in the context of quantum computation. ref. 31 demonstrates that AutoQEC can efficiently suppress noise during computation when the Knill-Laflamme condition is satisfied, by proposing methods for constructing unitary operations corresponding to logical gates. In contrast, we propose an ancilla-free AutoQEC code for quantum metrology, tailored to scenarios in which the unitary operation responsible for imprinting the signal is given. This highlights the essential distinction between AutoQEC in computation and in metrology, thereby consolidating the unique contribution of our work.

Methods

Proof of Theorem 1

Step 1.— Existence of code: First, we suggest a code that satisfies the Knill-Laflamme condition for the error set $\mathcal{E}^{[c]}$. Let us denote the vectors

satisfying the linear equation of (T2) as $\mathbf{p}_0 := ([p_0]_1, [p_0]_2, \dots, [p_0]_{N_0})^T$, $\mathbf{p}_1 := ([p_1]_1, [p_1]_2, \dots, [p_1]_{N_1})^T$. While multiple $\mathbf{p}_0 \in \mathbb{R}_{\geq 0}^{N_0}$, $\mathbf{p}_1 \in \mathbb{R}_{\geq 0}^{N_1}$ may exist satisfying the linear equation $\mathbf{A}_0 \cdot \mathbf{p}_0 = \mathbf{A}_1 \cdot \mathbf{p}_1$, our proof does not depend on the specific choice of $(\mathbf{p}_0, \mathbf{p}_1)$. Therefore, we consider an arbitrary $(\mathbf{p}_0, \mathbf{p}_1)$ that satisfies the homogeneous equation in (T2). Based on $(\mathbf{p}_0, \mathbf{p}_1)$, we consider the following two-dimensional code $\mathcal{C}$, with the codewords specified as follows:

$$|\mu_0\rangle := \sum_{k=1}^{N_0} \sqrt{[p_0]_k} |h_0^{(k)}\rangle, \quad |\mu_1\rangle := \sum_{k=1}^{N_1} \sqrt{[p_1]_k} |h_1^{(k)}\rangle. \quad (27)$$

Using Eqs. (9)–(11), one can easily verify that the existence of $(\mathbf{p}_0, \mathbf{p}_1)$ implies $\langle \mu_0 | \hat{E}_a^\dagger \hat{E}_b | \mu_0 \rangle = \langle \mu_1 | \hat{E}_a^\dagger \hat{E}_b | \mu_1 \rangle \forall \hat{E}_a, \hat{E}_b \in \mathcal{E}^{[c]}$. In addition, the first condition $[\hat{H}, \hat{L}_{n,i}] = 0 \forall i$ ensures that $\hat{E}_a^\dagger \hat{E}_b \forall \hat{E}_a, \hat{E}_b \in \mathcal{E}^{[c]}$ and $\hat{H}$ commute, resulting $\langle \mu_0 | \hat{E}_a^\dagger \hat{E}_b | \mu_1 \rangle = \langle \mu_1 | \hat{E}_a^\dagger \hat{E}_b | \mu_0 \rangle = 0 \forall \hat{E}_a, \hat{E}_b \in \mathcal{E}^{[c]}$. This can be easily proven using the following equation:

$$\begin{aligned} \langle \mu_0 | \hat{H} \hat{E}_a^\dagger \hat{E}_b | \mu_1 \rangle &= h_0 \langle \mu_0 | \hat{E}_a^\dagger \hat{E}_b | \mu_1 \rangle = \langle \mu_0 | \hat{E}_a^\dagger \hat{E}_b \hat{H} | \mu_1 \rangle \\ &= h_1 \langle \mu_0 | \hat{E}_a^\dagger \hat{E}_b | \mu_1 \rangle \forall \hat{E}_a, \hat{E}_b \in \mathcal{E}^{[c]}. \end{aligned} \quad (28)$$

Here, since $h_0 \neq h_1$, Eq. (28) is valid if and only if $\langle \mu_0 | \hat{E}_a^\dagger \hat{E}_b | \mu_1 \rangle = 0 \forall \hat{E}_a, \hat{E}_b \in \mathcal{E}^{[c]}$. As a consequence, we conclude that a code exists that satisfies the Knill-Laflamme condition for the error set $\mathcal{E}^{[c]}$ if the (T1) and (T2) in Theorem. 1 are satisfied.

Step 2.— Expression of Hamiltonian: For the second step of the proof, we note that $|\mu_0\rangle$ and $|\mu_1\rangle$ are eigenvectors of $\hat{H}$ with eigenvalues h_0 and h_1 , respectively. Furthermore, according to the definition of the eigenbasis of the correctable error spaces, the eigenbasis $|\mu_{i,i_n}^{[n]}\rangle$ (where $i \in \{0, 1\}$ and $1 \leq n \leq c$), can be expressed as a linear combination of $\{\hat{E}_a |\mu_i\rangle\}_a$ where $\hat{E}_a \in \mathcal{E}^{[c]}$. Combining this with $[\hat{H}, \hat{L}_{n,i}] = 0, \forall i$, it follows that $\hat{H} |\mu_{i,i_n}^{[n]}\rangle = h_i |\mu_{i,i_n}^{[n]}\rangle$ for $i \in \{0, 1\}$. These relations imply the following key orthogonality conditions:

$$\langle \mu_{i,i_n}^{[n]} | \hat{H} | \mu_{j,j_m}^{[m]} \rangle = h_i \delta_{i,j} \delta_{n,m} \delta_{i_n,j_m}, \quad (29)$$

$$\langle \phi_q | \hat{H} | \mu_{j,j_m}^{[m]} \rangle = \langle \phi_q | h_j | \mu_{j,j_m}^{[m]} \rangle = h_j \langle \phi_q | \mu_{j,j_m}^{[m]} \rangle = 0. \quad (30)$$

As a consequence, the Hamiltonian $\hat{H}$ can be expressed as

$$\begin{aligned} \hat{H} &= \sum_{j=0}^1 \sum_{n=0}^c \sum_{i_n=1}^{p_n} h_j |\mu_{j,i_n}^{[n]}\rangle \langle \mu_{j,i_n}^{[n]}| + \hat{\Pi}_{\text{res}} \hat{H} \hat{\Pi}_{\text{res}} \\ &= \sum_{j,k=0}^1 \sum_{n=0}^c \sum_{i_n=1}^{p_n} \langle \mu_j | \hat{H}_0 | \mu_k \rangle |\mu_{j,i_n}^{[n]}\rangle \langle \mu_{k,i_n}^{[n]}| + \hat{\Pi}_{\text{res}} \hat{H} \hat{\Pi}_{\text{res}}, \end{aligned} \quad (31)$$

where $\hat{\Pi}_{\text{res}} := \sum_{q=1}^{q_{\max}} |\phi_q\rangle \langle \phi_q|$ is the projector of the residual space, and $\hat{H}_0 := \sum_{j=0}^1 h_j |\mu_j\rangle \langle \mu_j|$.

To investigate whether there exists an engineered dissipation that achieves AutoQEC up to order c for quantum metrology when (T1) and (T2) are satisfied, let us revisit Lemma 1 in the main text which is the preliminary result of ref. 31. In Lemma 1, they state that if the Knill-Laflamme condition is satisfied for the error set $\mathcal{E}^{[c]}$, the AutoQEC up to order c can be achieved by applying the following engineered dissipation operation and the Hamiltonian

$$\tilde{\mathcal{L}}_E = \sum_{n=1}^c \sum_{i_n=1}^{p_n} D[\hat{L}_{E,i_n}^{[n]}] + \sum_{q=1}^{q_{\max}} D[\hat{L}_{E,q}^{[\text{res}]}], \quad (32)$$

$$\hat{H} = \sum_{j,k=0}^{d_c-1} \sum_{n=0}^c \sum_{i_n=1}^{p_n} \langle \mu_j | \hat{H}_0 | \mu_k \rangle | \mu_{j,i_n}^{[n]} \rangle \langle \mu_{k,i_n}^{[n]} |, \quad (33)$$

where $\hat{L}_{E,i_n}^{[n]} = \sum_{j=0}^{d_c-1} | \mu_j \rangle \langle \mu_{j,i_n}^{[n]} |$ for $1 \leq n \leq c$ and $1 \leq i_n \leq p_n$, and $\hat{L}_{E,q}^{[res]} = | \Phi_q \rangle \langle \Phi_q |$ for $1 \leq q \leq q_{\max}$. Here, $| \Phi_q \rangle$ can be chosen among any normalized states in $\mathcal{C}$. Notably, the only difference between the Hamiltonian designed for AutoQEC in computation (which is Eq. (33)) and the given signal Hamiltonian (which is Eq. (31)) is the term $\hat{\Pi}_{\text{res}} \hat{H} \hat{\Pi}_{\text{res}}$. This term does not affect the performance of AutoQEC since it operates exclusively within the residual space $\mathcal{R}$. As a consequence, by applying the engineered dissipation defined in Eq. (32), AutoQEC up to order c can be performed. Therefore, AutoQEC state $\hat{\rho}(t)$ satisfies

$$\| \tilde{\mathcal{P}}_E[\hat{\rho}(t)] - \hat{\rho}_{\text{id}}(t) \| \leq M \| \hat{\rho}(0) \| \frac{\kappa t}{R^c} \quad (34)$$

for all $t, R \geq 0$, where $\hat{\rho}_{\text{id}}(t) = \hat{U}_0(wt) \hat{\rho}_0 \hat{U}_0^\dagger(wt)$ with $\hat{U}_0(wt) = e^{-iwt\hat{H}_0}$ and $\hat{\rho}_0$ is the initial quantum probe.

Step 3.— QFI: Until now, we have demonstrated that if the sufficient condition in Theorem 1 is satisfied, AutoQEC up to order c can be performed. Lastly, we show that if AutoQEC up to order c is achieved, then, the QFI of the AutoQEC state achieves HS.

We note that AutoQEC up to order c is characterized by the operator norm of the difference between two states. Let us consider two states $\hat{\rho}$ and $\hat{\sigma}$ for further analysis. The difference of the two states can be expressed as

$$\hat{\rho} - \hat{\sigma} = \sum_i \lambda_i | \lambda_i \rangle \langle \lambda_i |, \quad (35)$$

where $| \lambda_i \rangle$ are the eigenvectors of $\hat{\rho} - \hat{\sigma}$ and the eigenvalues λ_i are ordered such that $| \lambda_i | \geq | \lambda_{i+1} |$. The operator norm of $\hat{\rho} - \hat{\sigma}$ is then given by $\| \hat{\rho} - \hat{\sigma} \| = | \lambda_1 | := x$. Since $x \geq | \lambda_i | \geq | \lambda_{i+1} |$, $\hat{\rho} - \hat{\sigma}$ can be expressed as

$$\hat{\rho} - \hat{\sigma} = \sum_{k=1}^{\infty} \hat{M}_k x^k. \quad (36)$$

As a consequence, the corresponding QFI difference can also be expressed as

$$F[\hat{\rho}] - F[\hat{\sigma}] = \sum_{k=1}^{\infty} F_k x^k. \quad (37)$$

Here, we note that since $\sum_{k=1}^{\infty} F_k x^k|_{x=0} = 0$,

$$\exists M_0 > 0, \exists x_0 \forall 0 \leq x < x_0 : \left| \sum_{k=1}^{\infty} F_k x^k \right| \leq M_0 x. \quad (38)$$

Next, let us consider $\hat{\rho} = \tilde{\mathcal{P}}_E[\hat{\rho}(t)]$ and $\hat{\sigma} = \hat{\rho}_{\text{id}}(t)$. We again emphasize that Eq. (38) holds for $x < x_0$ and according to Eq. (34), we have $x \leq M \| \hat{\rho}(0) \| \frac{\kappa t}{R^c} \leq M \| \hat{\rho}(0) \| \frac{\kappa T}{R^c}$. Therefore, for R satisfying $M \| \hat{\rho}(0) \| \frac{\kappa T}{R^c} < x_0$, equivalently $(M \| \hat{\rho}(0) \| \kappa T / x_0)^c < R$, we have

$$\begin{aligned} & | F[\tilde{\mathcal{P}}_E[\hat{\rho}(t)]] - F[\hat{\rho}_{\text{id}}(t)] | \\ & \leq M_0 M \| \hat{\rho}(0) \| \frac{\kappa t}{R^c} \leq M_0 M \| \hat{\rho}(0) \| \frac{\kappa T}{R^c}. \end{aligned} \quad (39)$$

For a given arbitrarily small error $\epsilon > 0$, by choosing R such that $(M_0 M \| \hat{\rho}(0) \| \kappa T / \epsilon)^c = R$, we have

$$| F[\tilde{\mathcal{P}}_E[\hat{\rho}(t)]] - F[\hat{\rho}_{\text{id}}(t)] | \leq M_0 M \| \hat{\rho}(0) \| \frac{\kappa T}{R^c} = \epsilon. \quad (40)$$

Next, we derive Eq. (12) from Eq. (40). In this derivation, we consider the scenario in which the QFI of the AutoQEC state is always less than or equal

to that of the ideal case over the entire sensing duration, i.e., $F[\hat{\rho}(t)] \leq F[\hat{\rho}_{\text{id}}(t)]$ for all $0 \leq t \leq T$. (We note that in cases where there exists some t such that $F[\hat{\rho}(t)] > F[\hat{\rho}_{\text{id}}(t)]$, the derivation remains straightforward.) In this case, since $\tilde{\mathcal{P}}_E$ is a well-defined CPTP map, the data processing inequality of the QFI ensures the following inequality holds⁴⁸:

$$F[\tilde{\mathcal{P}}_E[\hat{\rho}(t)]] \leq F[\hat{\rho}(t)] \leq F[\hat{\rho}_{\text{id}}(t)]. \quad (41)$$

We then obtain the following equation which can be easily derived from Eqs. (40) and (41):

$$F[\hat{\rho}_{\text{id}}(t)] - F[\hat{\rho}(t)] \leq \epsilon. \quad (42)$$

To proceed, we consider the initial quantum probe state $| \mu_0 \rangle + | \mu_1 \rangle$, for which the QFI of the ideal state is given by $F[\hat{\rho}_{\text{id}}(t)] = (h_0 - h_1)^2 t^2$. Consequently, substituting this expression into Eq. (42) leads directly to Eq. (12).

Data availability

The datasets generated and analyzed during the current study are not publicly available due to ongoing analyses and the inclusion of unpublished follow-up work, but are available from the corresponding author on reasonable request. The codes generated and analyzed during the current study are not publicly available due to ongoing analyses and the inclusion of unpublished follow-up work, but are available from the corresponding author on reasonable request.

Code availability

The codes generated and analyzed during the current study are not publicly available due to ongoing analyses and the inclusion of unpublished follow-up work, but are available from the corresponding author on reasonable request.

Received: 30 November 2025; Accepted: 3 May 2026;

Published online: 15 June 2026

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Acknowledgements

We acknowledge useful discussions with Ivan Rojko, Sisi Zhou and Kyungjoo Noh. This research was funded by National Research Foundation of Korea (RS-2022-NR068812, RS-2024-00442762), Institute of Information & Communications Technology Planning & Evaluation (RS-2025-02263264), and Global Partnership Program of Leading Universities in Quantum Science and Technology (RS-2025-02317602). H.K. is supported by the Education and Training Program of the Quantum Information Research Support Center (No.~2021M3H3A1036573), the IITP (RS-2025-25464252, RS-2024-00437191), the NRF (RS-2025-25464492, RS-2024-00442710) funded by the Ministry of Science and ICT (MSIT) Korea. U.R.F. was supported by the NRF under Grant no.~2020R1A2C2008103. L.J. acknowledges support from the ARO(W911NF-23-1-0077), ARO MURI (W911NF-21-1-0325), AFOSR MURI (FA9550-19-1-0399, FA9550-21-1-0209, FA9550-23-1-0338), DARPA (HR0011-24-9-0359, HR0011-24-9-0361), NSF (OMA-1936118, ERC-1941583, OMA-2137642, OSI-2326767, CCF-2312755), NTT Research, and the Packard Foundation (2020-71479).

Author contributions

H.K. conceived the idea of applying autonomous quantum error correction to quantum metrology, performed all theoretical analyses and numerical simulations, prepared the figures, wrote the manuscript, and revised the manuscript through regular discussions with L.J. and S.-W.L. U.R.F. conceived the main idea of applying autonomous quantum error correction to achieve temporal Heisenberg scaling in quantum metrology and revised the manuscript. S.-W.L. supervised the project, contributed to the theoretical development of the theorems, identified relevant examples and implementation platforms, and revised the manuscript. L.J. conceived the idea of applying autonomous quantum error correction to quantum metrology, supervised the project, contributed to the theoretical development of key results and relevant examples, provided overall guidance and feedback, and revised the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at

<https://doi.org/10.1038/s41534-026-01268-1>.

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